Deepthi Venmanasseril Jainlal

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Education

MS in Engineering Management (GPA: 4.0) — **Westcliff University** — *California, USA* — **10/2025 – 10/2027**

- **Coursework**: Project Management, Big Data Analytics, Business Analytics

MTech in Computer Science — **VIT University** — *Vellore, India* — **08/2019 - 07/2021**

[Specialization in Artificial Intelligence and Machine Learning]

- **Coursework**: Algorithms, Data Structures, Big Data, Databases, Inference, Statistics, Artificial Intelligence, Machine Learning, RL, Statistics.

Professional Experience (EXPERIENCE: 4.6 YEARS)

Ford Motors

Data Scientist – Supply Chain Analytics | Inventory Planning & Optimization | Predictive Modeling | Cost Analysis **08/2023 - 07/2025**

- Analyzed **supplier pricing, part costs, and commercial adjustments** to support **inventory cost optimization and product availability decisions** using **Qlik Sense, Alteryx, Dataform, and BigQuery**. Also built and maintained **ETL pipelines** on **GCP**
- Designed a Python-based LP model to optimize consolidation center selection, improving **inventory flow, cost, and planning efficiency.**
- Developed **Lane Rate estimation dashboard and Consolidation Center Selection Optimization using Gurobi (Inventory Flow & Logistics Cost Reduction)** using machine learning algorithms and data engineering with **Alteryx, GCP, and Dataform to support production planning, logistics optimization**.
- Created a dashboard to identify **amortization opportunities in suppliers**, resulting in **$23 million in potential savings** by leveraging **Gemini LLM and prompt engineering**; recognized within the team for this project, improving **cost, quality, and supplier compliance.**
- Developed **Nuts and Bolts Price Estimation Dashboard** using **Random Forest algorithm and Power BI**, achieving **$5 million savings**.
- Received **Recognition for Excellence** and **Recognition for Collaboration** from Ford for improving Lane Rate value accuracy and contributions to cross-functional teams.
- Mentored junior engineers on ML model development, optimization techniques, and production deployment best practices.

PHILIPS HEALTHCARE

Software Engineer (also Machine learning Engineer as GIG) **07/2021 - 07/2023**

- Developed features for **patient monitor support software** using **C#, .NET, and gRPC**.
- Removed redundant code and replaced **Perl scripts with C#**, enhancing performance of the patient monitor upgrade by **25%**.
- Designed and implemented a **test automation project** for the support tool application, reducing manual testing time by **60%**.
- Developed a **patient monitor simulator** enabling easy testing of new web-based support tool versions, reducing testing complexity by **80%**; simulated capabilities of various patient monitors not present locally.
- Developed **risk analysis strategies** using machine learning algorithms and statistical methods including **Monte Carlo simulation**; presented results to the risk team to help understand and analyze risk patterns and features.
- Generated **synthetic data** using machine learning algorithms and forecasted risk analysis metrics using **Random Forest algorithm**.

Deep Learning Research Intern **09/2020 - 07/2021**

- Developed **deep learning models** for medical image segmentation using **CNNs (YOLOv3, nnU-Net)** and **PyTorch**, achieving **89% accuracy** (large-scale model training).
- Annotated **900 ultrasound images(clinical imaging)** and created masks using **Python and OpenCV** in close collaboration with doctors.

Projects

- **Agentic AI Used car Inspection Agent**: Developed an autonomous agent leveraging Computer Vision and LLMs to automate secondhand car inspections (Google AI Intensive); integrated Vision-Language Models (VLM) for real-time defect detection.
- **Real-Time Emotion Classifier**: Built a high-speed facial expression recognition system using CNNs, PyTorch, and OpenCV, capable of classifying 7 core emotions in live video streams.
- **Raspberry Pi telegram Chatbot**: Developed a chatbot integrating NLP intents and the Telegram API.
- **Consolidation center selection Optimization:** Designed a Python-based LP optimization model using PuLP to maximize ad conversions under budget constraints, with solver logic transferable to Gurobi/CPLEX

Blogs/Publications

- Solving Real Logistics Problems: Building a Consolidation Center Optimizer in Python
- Evaluating Sampling Efficiency in Bayesian Inference: A Benchmark of Markov Chain Monte Carlo vs. Hamiltonian Monte Carlo
- The Ultimate Guide: Hybrid Anomaly Detection with Autoencoders & XGBoost.

Core Competencies

- **Languages and tools :** Python | R programming | C# | C | SQL | BigQuery
- **Libraries & Frameworks & Tools :** PyTorch | TensorFlow | Scikit-learn | Pandas | NumPy | Matplotlib | NLTK | Gurobi | Transformers | LLM | Agentic AI | Git | Jira (Agile)| CI/CD Tools | Jupyter | Docker | MySQL | qlik sense, Power BI, Alteryx| GCP | Statistical Computing | Generative AI | Reinforcement Learning